

# DG4RAS 25/26 Assessment Brief: Project

## 1.0 Essential Information

|                          |                                                                                                                                                                                                                                                           |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module Code and Title    | DG4RAS – Robotics and Autonomous Systems                                                                                                                                                                                                                  |
| Assessment Contact       | Luis J. Manso <l.manso@aston.ac.uk>                                                                                                                                                                                                                       |
| Assessment Title         | Project                                                                                                                                                                                                                                                   |
| Assessment Weighting     | 75%                                                                                                                                                                                                                                                       |
| Submission Date and Time | The project is composed of 3 missions. Each of which has its own deadline, as follows:<br>3 <sup>rd</sup> November 2025 at 16:30 (Mission 1)<br>20 <sup>th</sup> November 2025 at 16:30 (Mission 2)<br>8 <sup>th</sup> December 2025 at 16:30 (Mission 3) |
| Word Count               | Not applicable.                                                                                                                                                                                                                                           |

## 2.0 Assessment Overview

### Assessment Mode

You will need to produce Python code to control robots simulated using Webots R2023b in a series of three missions covering different units of the module:

1. Perception and behaviour coordination for a mobile robot.
2. Navigation with reactive obstacle avoidance.
3. Control of an autonomous manipulator for a pick-and-place task.

With each mission you can get up to 25 marks, adding up to 75 marks in total. They will be assessed with the help of a 30-minute face-to-face demo, in person. If you find anything in the brief that is not clear, please let the module tutors know, so that it can be clarified.

### Task Description

To design and develop all missions, and to demonstrate them in a live face-to-face demo in which you will explain the design and implementation of your control system. If you don't book or attend a demo you will receive zero marks for this assessment, unless you are granted Exceptional Circumstances. You will be able to book your demo within the 5 working days following the last submission.

With each mission, you will be given a template project which includes the simulation and code you can build on top of to carry out your task: to design and implement a controller to achieve the desired behaviours. You are not allowed to change the simulation file, but you can add new Python modules that your controller can use and load 3rd party libraries (if referenced appropriately).

#### Mission 1: Basic Perception, Control, and Behaviour Coordination (25 marks)

- Scenario: Controlling a robot racing in a simulated track. The robot is equipped with an RGB camera, a 2-D LiDAR, a GPS, and a compass. The environment will be like the one used in the second week's tutorial, but with obstacles in the track.
- Desired behaviour: The robot should do as many laps as possible, as fast as possible, without colliding with the obstacles or leaving the track.
- Code comments should provide a rationale for how you implemented perception, control and behaviour coordination.

#### Mission 2: Navigation with reactive obstacle avoidance (25 marks)

- Scenario: You will be in control of a robot guide in a museum. Your robot will be asked to guide a person toward specific pieces of art. Once your robot is assigned guiding the person to a specific piece of art, the person will follow your robot. Unfortunately, the person is afraid of robots and will run away if other robots get close. Therefore, you will need to guide the person while avoiding getting close to other robots. The tasks are given using the keyboard. New tasks may come up at any point in time and should be queued.
- Desired behaviour: The robot should **purposefully** navigate to the target avoiding distressing the

museum's client. The robot should also avoid any static or moving obstacles. Upon fulfilling its tasks, it should be ready to take further instructions.

- Your code comments should provide a rationale for how you solve obstacle avoidance, local navigation, global navigation, and your behaviour coordination strategy.

### **Mission 3: Pick-and-place with an autonomous manipulator (25 marks)**

- Scenario: A two-armed robot equipped with cameras is tasked with cleaning the table. The existing controller provides functions to move the robot arms in joint and task space, and to operate the gripper. There are several objects on the table and a bin.
- Desired behaviour: The robot should detect the objects on the table using its cameras and drop them one by one in the bin, until the table is empty. The bin will always be in the same position.
- In your code comments, you should provide a rationale for how you solve navigation, object detection and manipulation.

For each mission, there will be commented code in the template provided. You are only allowed to modify the controller of your robot, and not any other files that already exist. Note that accomplishing the desired behaviours is only part of the mark, both the **rationale** and the documentation within the code count as well. See marking rubric at the end of this document.

Please bear in mind that any documentation should be included as code comments (not as a separate document), and that you will demonstrate the exact same version that you submitted.

Use of machine learning –classical or otherwise – is welcome but not mandatory, provided that you have found other appropriate means to achieve the mission.

### **Submission Instructions (When and Where do I submit?)**

Please submit your assessment via your module's Assessment Area on Blackboard. Guidance on how to submit can be found in the following link: [guide to submitting to Blackboard assignments](#).

Each mission must be submitted as a compressed ZIP folder containing the entire project. If you make use of any deep learning models, the models do not necessarily have to be included, as long as they are automatically downloaded (or a URL is clearly provided in the code). As aforementioned, if you don't book an appointment, your project will receive zero marks. Lateness will carry a 10% penalty per day, up to a week, and it applies to each mission separately based on the respective submission deadline. If you think you have grounds to apply for Exceptional Circumstances (EC), please read the following link: [Academic Support](#).

Feel free to use existing Python packages. Whenever you use or copy code from third-party resources, you MUST EXPLICITLY STATE in the documented code what pieces of code have been copied and where the code has been copied from. If you fail to do so, we may not be able to give marks for writing such code and you may be reported for academic malpractice.

## **3.0 Purpose of Assessment**

### **Link to Learning Outcomes**

The project covers the Learning Outcomes 1-3 from the module specification:

1. **Apply perception algorithms to extract information from sensors:** You will learn to assess which are the most appropriate algorithms to process sensor data in a way that it is useful for controlling robots, taking the application and the robot itself into consideration, and putting the theory and tutorials into practice.
2. **Process sensor data to recognise objects in the environment and navigate:** In the project, percepts will be used to navigate the environment and to enable robot manipulation.
3. **Design a control method for a robotic system in a given context by selecting and composing sensing, classical machine learning and decision-making algorithms:** In autonomous robots, perceiving and modelling the environment only makes sense in the context of decision making. You will design your own robot control systems.

### Purpose of Assessment

The rationale behind this task is to create an opportunity for you to align and connect what we have explored in the module – practical elements mostly, but also theoretical ones. The three missions consider different types of robots in different contexts and will help you get a breadth of experience in the main areas of autonomous robots. You will develop technical skills on the design of software for perception and control of autonomous robots, and critical thinking applied to problem solving.

## 4.0 Feedback Expectations

### When and where can I find my grades and feedback?

You will receive formative and summative feedback within 4 weeks of the submission deadline of the third mission. For each mission, you will receive feedback on every aspect included in the marking rubric (see marking rubric below). The feedback will be accessible via Blackboard and will include a provisional mark. Any grades that you see outside of [My Aston Portal \(MAP\)](#) should be thought of as provisional. Only final Exam Board approved grades will appear in MAP. Feedback and provisional grades can be accessed in Blackboard. You can read this [Grades Guide](#).

## 5.0 Guidance on AI Usage

### Can I use AI in my work and how?

#### Use of Gen AI is Optional



It is acceptable to make constructive use of generative AI tools for this assignment. The purpose of the task is to demonstrate your skills in designing and developing control software for autonomous robots. This means that you can use Generative AI to write small excerpts of code, but not to design the code for any of the missions. If you use Generative AI to design your code, you will most likely struggle to explain your design decisions and how the code works. Any application or use of AI should always be acknowledged.

Misuse of generative AI, as defined in Aston's assessment regulations, will be treated as an academic offence and may be subject to disciplinary action. You may find this guidance on the effective and ethical use of AI for assessment useful: [Artificial Intelligence for study and assessment | Rise 360](#)

## 6.0 Assessment Tips

### What further information and advice might help me to complete this task?

- Make sure you create regular backups of your missions. Using Git is highly recommended.
- Include in your submission anything you may have implemented. Even if it doesn't work, you can still get some marks for it!
- To make the demo easier for you, you can include justifications of your design choices and how they relate to the module content or additional readings in your code comments, as this directly relates to the rationale component of the marking scheme and will be discussed in the demos.
- Make sure that you have not modified the world file of the simulation. If you did, overwrite the world file (**not the code**) with the one in the template.
- The module tutors are happy to guide you. Make use of the office hours if you need help.

## 7.0 Assessment Criteria and Rubrics

### What will my work be marked against?

For each of the 3 missions you will get up to 25 marks. The demo is mandatory, and no marks will be given without a demo. You will get marks for the functionality that you **can** explain. The missions are marked independently within a single demonstration appointment. Each mission will be marked based on your rationale, the performance of the robot, and the documentation of your code, as follows:

| Rationale                                                      | 12  |
|----------------------------------------------------------------|-----|
| There is little or no evidence of principled design decisions. | 0-3 |

|                                                                                                                                                                                                                                                                                 |          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| There is some evidence of principled design decisions, but they fall below the expected threshold, or the student fails to explain the general design principles of their solution.                                                                                             | 4-5      |
| Adequate, principled design decisions have been made based on the content of the module and possibly further research. The general design principles of their solution were explained adequately. The behaviour is not hand-crafted for the specific layout of the environment. | 6-10     |
| The rationale is informed by the content of the module as well as further research. The student provided insightful explanation of the design principles.                                                                                                                       | 11-12    |
| <b>Performance</b>                                                                                                                                                                                                                                                              | <b>8</b> |
| The code does not run, the robot does not move, or there are other obvious flaws such as assuming the position of the obstacles or objects to interact with (this does not apply to objects that are generally expected to be static such as tables or walls).                  | 0-1      |
| The code works, but there are major flaws which prevent the robot from executing the desired behaviour. There are significant gaps in the explanations of the algorithms developed or their implementation.                                                                     | 2-3      |
| The implementation is generally sound, and the robot executes the desired behaviour with only minor errors or inconsistencies (if any).                                                                                                                                         | 4-7      |
| The behaviour of the robot is excellent over a variety of edge cases (e.g., including obstacles). The student can explain how their implementation works in detail.                                                                                                             | 8-9      |
| <b>Documentation</b>                                                                                                                                                                                                                                                            | <b>5</b> |
| There are some comments in the code, but they fall short of the expected documentation (e.g., comments apply to lines only, or are not effective in explaining the purpose of each piece of code).                                                                              | 0-1      |
| Most functions and classes have been documented with their purpose, inputs, and outputs, and additional comments throughout help understanding the code. Relevant design decisions can be inferred from the comments.                                                           | 2-3      |
| All functions are documented with their purpose, inputs, and outputs following standard practices in code documentation. Additional comments throughout help understanding the code. The rationale and design decisions are clearly stated.                                     | 4-5      |